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25 August 1976
SUPERSEDING
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MILITARY SPECIFICATION

SODIUM BROMATE, ANALYZED REAGENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one grade of sodium bromate.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-301 - Chemicals, Dry and Paste: Packaging and Packing Of.

STANDARDS

FEDERAL

Fed. Std. No. 356 - Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6810

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

ASTM D1193 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Appearance. Sodium bromate shall be a colorless or white crystalline powder when tested as specified in 4.2.4.1.

3.2 Chemical characteristics. Sodium bromate shall conform to the chemical characteristics of table I when tested as specified therein.

TABLE I. Chemical characteristics

Characteristic	Percent by weight: Minimum	Maximum	Test paragraph
Assay (as $NaBrO_3$)	99.7	----	4.2.4.2
Insoluble matter	----	0.01	4.2.4.3
Free acid (as $HBrO_3$)	----	0.01	4.2.4.4
Free alkali	----	To pass test	4.2.4.5
Bromide	----	0.05	4.2.4.6
Sulfate	----	0.015	4.2.4.7
Heavy metals (as Pb)	----	0.0005	4.2.4.8
Iron	----	0.002	4.2.4.9

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the sodium bromate produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test (see 6.4). Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 100 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample containers shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified

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(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests (see 6.4). Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Appearance. Visually examine the specimen for form and color.

4.2.4.2 Assay. Powder approximately 0.5 g of specimen and dry to constant weight over sulfuric acid. Weigh to the nearest 0.1 milligram (mg) approximately 0.1 g of the dried specimen and dissolve in 50 milliliters (ml) of water in a glass-stoppered flask. Add 3 g of potassium iodide and 3 ml of 37-percent hydrochloric acid which has been previously diluted with 10 ml of water. Allow to stand for 5 minutes, then add 50 ml of cold water and titrate the liberated iodine with 0.1N sodium thiosulfate solution. Calculate the percent by weight sodium bromate as follows:

$$\text{Percent sodium bromate} = \frac{0.2515A}{W}$$

where: A = Milliliters of 0.1N sodium thiosulfate solution used in the titration and
W = Weight of specimen in grams.

4.2.4.3 Insoluble matter. Weigh to the nearest 0.1 g approximately 10 g of the specimen and dissolve in 100 ml of hot water. Heat on a steam bath for 1 hour. Filter any residue through a tared filter crucible, wash with hot water, and dry to constant weight (weighed to the nearest 0.1 mg) at 105° ± 3° C. Calculate the percent by weight insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100A}{W}$$

where: A = Weight of residue in grams and
W = Weight of specimen in grams.

4.2.4.4 Free acid. Dissolve 5.00 g of the specimen in 60 ml of warm water and add 3 drops of phenolphthalein indicator. Add 0.002N sodium hydroxide solution until a pink color is produced. No more than 2.0 ml of the sodium hydroxide solution shall be required to produce the pink color.

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4.2.4.5 Free alkali. Dissolve 5.00 g of the specimen in 60 ml of warm water and add 3 drops of phenolphthalein indicator. A pink color shall not be produced.

4.2.4.6 Bromide. Dissolve 4.00 g of the specimen in 80 ml of water. Add 0.25 ml of 1N sulfuric acid to 40 ml of the specimen solution. Allow to stand for 2 minutes. The color of the solution to which the sulfuric acid has been added shall be no darker than the remaining 40 ml of specimen solution.

4.2.4.7 Sulfate. Add 1.00 g of specimen and 0.05 g of sodium carbonate to 15 ml of water and 10 ml of 37-percent hydrochloric acid and evaporate to dryness. Take up the residue with 20 ml of water. Add 1 ml of 1N hydrochloric acid and 2 ml of 12-percent barium chloride solution. Prepare a control by evaporating 10 ml of 37-percent hydrochloric acid and 0.05 g of sodium carbonate to dryness on a steam bath. Dissolve the residue and 0.150 mg of sulfate (1.5 ml of a solution containing 0.1814 g of potassium sulfate in sufficient water to make 1 liter) in sufficient water to make 20 ml, then add 1 ml of 1N hydrochloric acid and 2 ml of 12-percent barium chloride solution. Any turbidity produced in the specimen solution shall be no darker than that produced in the control solution.

4.2.4.8 Heavy metals. Add 4.00 g of specimen to 20 ml of water and 15 ml of 37-percent hydrochloric acid and evaporate to dryness on a steam bath. Reevaporate with 10 ml of 37-percent hydrochloric acid. Take up the residue with water and dilute to 40 ml. To 10 ml of solution add 0.01 mg of lead (1 ml of a solution containing 0.01598 g of lead nitrate in sufficient water to make 1000 ml), dilute to 30 ml with water, and add 1 ml of 1N acetic acid (this is solution A). To the remaining 30 ml of solution, add 1 ml of 1N acetic acid (this is solution B). Add 10 ml of freshly-prepared hydrogen sulfide solution to each solution. The color of solution B shall be no darker than that of solution A. Correct for any heavy metals derived from the hydrochloric acid by evaporating 25 ml of the acid to dryness on a steam bath, then taking up the residue with water and treating with hydrogen sulfide.

4.2.4.9 Iron. Add 0.040 mg of iron (4 ml of a solution containing 0.0864 g of ferric ammonium sulfate in sufficient water to make 1000 ml) to solution A from 4.2.4.8 and make solution A and solution B from 4.2.4.8 alkaline with 28-percent ammonium hydroxide. The color of solution B shall be no darker than that of solution A.

5. PACKAGING

5.1 Packaging, packing, and marking. A 1/4 pound (+0.04 or -0 ounce) or a 1 pound (+0.16 or -0 ounce) quantity of sodium bromate as specified (see 6.2) shall be packaged, packed, and marked level A, level B, or commercial as specified (see 6.2).

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5.1.1 Levels A and B.

5.1.1.1 Packaging and packing. Packaging and packing shall be in accordance with the requirements for a type I, class 1 (enclosure a) bottle as specified in PPP-C-301.

5.1.1.2 Marking. Marking shall be in accordance with MIL-STD-129.

5.1.2 Commercial. Packaging, packing, and marking shall be as specified in Fed. Std. No. 356. Marking shall include the lot or batch number and date of manufacture.

6. NOTES

6.1 Intended use. Sodium bromate, analyzed reagent is intended for use as a laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Unit quantity required (see 5.1).
(c) Level of packaging, packing, and marking required (see 5.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Sampling and testing caution. When sodium bromate is rubbed with organic substances, sulfides, or readily oxidizable substances it may react with explosive violence. It should be stored in a dry place and containers should be kept tightly closed.

6.5 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of Recommended Practice for Indicating Which Places of Figures Are To Be Considered Significant in Specified Limiting Values (ASTM E29).

Custodians:

Army - EA
Air Force - 68

Review activities:

Army - MD
DSA - GS

Preparing activity:

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OMB Approval
No. 22-R255

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